

Problem 11.13

A savings and loan association pays 7% effective on deposits at the end of each year. At the end of every 3 years a 2% bonus is paid on the balance at that time. Find the effective rate of interest earned by an investor if the money is left on deposit

- (a) Two years.
- (b) Three years.
- (c) Four years.

Solution

Let $a(t)$ be the accumulated value of an initial deposit of 1. The equivalent annual effective interest rate over t years satisfies

$$(1 + i)^t = a(t), \quad i = a(t)^{1/t} - 1.$$

Each year contributes an accumulation factor of 1.07. The bonus at the end of year 3 multiplies the balance, including that year's interest, by 1.02.

Part (a): Two years.

No bonus has been paid by the end of year 2, so

$$a(2) = (1.07)^2.$$

Therefore,

$$(1 + i)^2 = (1.07)^2, \quad \boxed{i = 7\%}.$$

Part (b): Three years.

At the end of year 3, one bonus is paid. Thus,

$$a(3) = (1.07)^3(1.02).$$

The equivalent annual effective rate is

$$i = [(1.07)^3(1.02)]^{1/3} - 1 = 1.07(1.02)^{1/3} - 1.$$

Hence,

$$\boxed{i \approx 7.70862992\%}.$$

Part (c): Four years.

The balance, including the bonus paid at time 3, earns another year of interest. No additional bonus is paid at time 4, so

$$a(4) = (1.07)^4(1.02).$$

Therefore,

$$i = [(1.07)^4(1.02)]^{1/4} - 1 = 1.07(1.02)^{1/4} - 1.$$

Hence,

$$\boxed{i \approx 7.53103368\%}.$$